

EduGenie

Google Gemini Powered Learning Assistant

Phase 4 – Project Planning

Project Schedule, Team Organization, Resource Planning and Risk Management

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1. Introduction

The Project Planning phase is fundamental to establishing a structured and organized approach for developing the EduGenie AI-powered learning assistant. This phase defines how the project is organized and executed from requirements analysis through development, testing, documentation, and final demonstration. Effective project planning helps the team accomplish the following objectives:

- Organize project activities into logical phases and stages
- Divide the work into manageable, clearly defined deliverables
- Identify required resources, technologies, and infrastructure
- Establish project milestones and track progress effectively
- Anticipate and manage project risks proactively
- Coordinate team activities and ensure clear communication
- Maintain consistency and quality throughout the development lifecycle

2. Project Planning Overview

EduGenie is planned and structured as a modular AI-powered educational assistant designed to support learners through intelligent tutoring and content synthesis. The project planning process encompasses the following sequential stages: **Requirement Analysis** → **Project Design** → **Development** → **Integration** → **Testing** → **Documentation** → **Demonstration**. Each stage contributes systematically to building and validating the complete learning assistant. Requirement analysis identifies what the system must do and for whom. Project design defines the architecture and technical approach. Development implements the core modules and integrations. Testing validates functionality and quality. Documentation captures implementation details and usage. Demonstration showcases the working system to stakeholders.

3. Project Objectives

The EduGenie project aims to achieve the following objectives:

1. Develop an AI-powered learning assistant to support students across various academic levels
2. Provide accurate, concise answers to educational questions
3. Simplify difficult concepts through clear, understandable explanations
4. Generate quizzes for self-assessment and learning practice
5. Summarize educational content for efficient studying and revision
6. Provide personalized learning recommendations and stepwise guidance
7. Integrate Google Gemini 2.5 Flash for appropriate and effective generative AI functions
8. Provide a simple, accessible web-based interface for diverse users
9. Organize the project into modular, maintainable backend components
10. Develop a scalable, understandable, and maintainable educational application

4. Project Scope

4.1 In-Scope

The current project scope includes the following features and components:

- Educational Question Answering (Q&A; module)
- Concept Explanation (simplified learning explanations)
- Quiz Generation (multiple-choice question generation)
- Content Summarization (condensing long passages)
- Learning Recommendations (personalized learning paths)
- Web-based frontend interface (HTML, CSS, JavaScript)
- FastAPI backend with RESTful architecture
- Google Gemini 2.5 Flash API integration
- LaMini-Flan-T5-783M local model for concept explanation
- API-based communication between frontend and backend

4.2 Out-of-Scope / Future Enhancements

The following features are identified as possible enhancements for future versions and are not part of the current project scope:

- Voice interaction and speech recognition
- Multilingual support and internationalization
- Mobile applications (iOS/Android)
- Offline functionality and local-first features
- Learning progress dashboards and analytics
- Gamification elements (badges, streaks, rewards)
- Adaptive learning paths based on user performance
- Group study and collaborative features
- Teacher and parent dashboards
- Learning Management System (LMS) integration
- Image and PDF input processing
- Real-time synchronization across devices
- Push notifications and alerts

5. Team Organization

The EduGenie project is organized around a dedicated team of five members with clear roles and responsibilities:

Name	Role
Rithik Sharan	Team Lead
Sakti Murugan	Team Member
Nivash	Team Member
Shreya Bharat	Team Member
Sri Niranjani	Team Member

Team Lead Responsibilities: The Team Lead (Rithik Sharan) coordinates overall project activities, manages communication between team members, oversees planning and resource allocation, and ensures progress towards project milestones. **Team Members:** Team members contribute to project activities including development, testing, documentation, code review, and other tasks according to the team's requirements and work allocation.

6. Work Breakdown Structure

The Work Breakdown Structure (WBS) organizes the entire project into manageable components and activities:

1. Project Initiation

- Project idea identification
- Problem identification
- Brainstorming and ideation

2. Requirement Analysis

- Identify target users
- Identify functional requirements
- Identify non-functional requirements
- Identify technical requirements

3. Project Design

- System architecture design
- Workflow design
- Module design
- Frontend design
- Backend design
- API design

4. Project Development

- Environment setup
- Backend development
- Frontend development
- Q&A; module development
- Explanation module development
- Quiz module development
- Summary module development
- Learning recommendation module development
- Gemini 2.5 Flash integration

- Local model integration
- Module integration

5. Testing

- Functional testing
- API testing
- Module testing
- Integration testing
- Bug identification and correction

6. Documentation

- Project documentation
- Phase documentation
- Technical documentation
- User/project explanation documentation

7. Demonstration

- Project execution
- Feature demonstration
- Working-process explanation
- Final output demonstration
- Demo video preparation

7. Project Activities

Major project activities are organized by development stage:

Stage	Major Activities
Initiation	Brainstorming and problem identification
Requirement Analysis	Requirements and scope identification
Design	Architecture, workflow, module and interface design
Development	Frontend, backend and AI module development
AI Integration	Gemini 2.5 Flash and local model integration
Testing	Functional, API and integration testing
Documentation	Technical and project documentation
Demonstration	Final execution and project demonstration

8. Technology and Resource Planning

8.1 Software Resources

The following software resources and tools are used in the EduGenie project:

- Python (version 3.10+) – Primary programming language
- FastAPI – Modern web framework for building RESTful APIs
- HTML5 – Frontend markup language
- CSS3 – Frontend styling and responsive design
- JavaScript – Frontend interactivity
- Google Gemini 2.5 Flash API – Cloud-based generative AI
- LaMini-Flan-T5-783M – Local instruction-tuned language model
- Uvicorn – ASGI server for running FastAPI applications
- Jinja2 – HTML templating engine
- Python packages – Required dependencies and libraries
- Git/GitHub – Version control and source code management

8.2 Hardware Resources

The project requires a computer capable of running the Python FastAPI application with all required dependencies and the local LaMini-Flan-T5-783M language model. The system should have sufficient processing power and memory to handle concurrent requests and model inference operations.

8.3 Internet Requirements

Internet connectivity is required for API functionality that communicates with the Google Gemini 2.5 Flash API. The application must maintain a stable connection to cloud services for AI-powered features.

9. Project Timeline

The following represents the planned project timeline, organized into eight sequential development phases. This timeline indicates the planned progression through the project lifecycle:

Phase	Planned Activity
Phase 1	Brainstorming and Ideation
Phase 2	Requirement Analysis
Phase 3	Project Design
Phase 4	Project Planning
Phase 5	Project Development
Phase 6	Project Testing
Phase 7	Project Documentation
Phase 8	Project Demonstration

Planned Project Execution Timeline

Phase 1 → Phase 2 → Phase 3 → Phase 4 → Phase 5 → Phase 6 → Phase 7 → Phase 8
(Ideation) → (Analysis) → (Design) → (Planning) → (Development) → (Testing) → (Documentation) → (Demonstration)

10. Project Milestones

Project milestones mark significant achievements and decision points throughout the development lifecycle:

Milestone	Description
M1	Project idea and problem identified
M2	Requirements defined and approved
M3	System design completed
M4	Project plan established
M5	Core application developed
M6	AI and module integration completed
M7	Testing and corrections completed
M8	Documentation completed
M9	Final project demonstration prepared

11. Risk Management

Effective risk management involves identifying potential challenges and planning mitigation strategies:

Risk	Possible Impact	Mitigation
AI-generated responses may contain inaccurate or sensitive information	Confidentiality issues	Review and validate important outputs
Gemini API availability or configuration errors	Application may not work correctly	Maintain correct API configuration and environment setup
API key exposure	Security risk	Store credentials in environment variables and never commit secrets
Local model resource requirements	Explanation module may require sufficient system resources	Ensure the development environment can run the local model
Prompt quality	Output quality may vary	Refine and test prompts iteratively
Integration issues	Modules may not work together correctly	Perform module and integration testing
Frontend/backend communication issues	Requests may fail or results may not display	Test API endpoints and frontend interaction thoroughly
Documentation inconsistencies	Project submission may become unclear	Review all phase documents for consistency

12. Dependency Planning

Project dependencies identify relationships and prerequisites between different components:

Frontend Dependencies: The frontend web interface depends on the backend API endpoints for processing requests and returning results.

Module Dependencies: Learning modules depend on their configured AI/model components (Gemini 2.5 Flash or local models) functioning correctly.

API Dependencies: Gemini-based features depend on valid API credentials, configuration, and stable internet connectivity.

Model Dependencies: The explanation module depends on the LaMini-Flan-T5-783M model and all required environment setup.

Testing Dependencies: Testing activities depend on the availability of developed modules and the integrated system.

Documentation Dependencies: Final documentation and demonstration depend on the completed and functioning project.

Overall Dependency Flow: Requirements → Design → Development → Integration → Testing → Documentation → Demonstration

13. Quality Planning

Quality assurance ensures that EduGenie meets defined standards across all aspects of development:

Code Quality

- Maintain organized module structure and clear separation of concerns
- Write readable, well-commented code with consistent naming conventions
- Separate different learning functions into appropriate modules
- Follow best practices in API design and implementation

Functional Quality

- Verify that each learning function performs its intended purpose
- Test API endpoints for correct request/response handling
- Verify frontend/backend interaction and data flow
- Ensure proper error handling and user feedback

AI Output Quality

- Review AI-generated educational responses for accuracy and relevance
- Refine prompts iteratively based on output quality assessment
- Check outputs for clarity, appropriateness, and educational value

Documentation Quality

- Maintain consistency between project implementation and documentation
- Avoid claiming features that are not implemented
- Keep all eight SmartBridge phases organized and documented
- Ensure clarity and completeness in technical explanations

14. Project Deliverables

Project deliverables are organized according to the eight SmartBridge methodology phases:

Phase	Deliverable
Phase 1	Brainstorming and Ideation documentation
Phase 2	Requirement Analysis documentation
Phase 3	Project Design documentation
Phase 4	Project Planning documentation
Phase 5	Project source code and development documentation
Phase 6	Testing documentation and test evidence
Phase 7	Complete Project Documentation
Phase 8	Project Demonstration and demo evidence

The project repository should organize all work according to these eight phases, with clear documentation for each milestone. Each phase builds upon previous phases to create a comprehensive, well-documented

educational software project.

15. Project Monitoring and Tracking

Project progress can be monitored and tracked by observing the following key indicators:

- Completion of planned activities according to the Work Breakdown Structure
- Development and functionality of individual modules
- Progress on module integration and system integration
- Testing progress and bug resolution rate
- Documentation completeness and consistency
- Resolution of identified issues and pending tasks
- Milestone achievement and on-time delivery
- Quality of deliverables against defined standards

The team should periodically review project status, identify incomplete tasks, resolve emerging issues, and adjust the plan if necessary. Regular status reviews help maintain project momentum and ensure successful delivery.

16. Team Coordination

Effective team coordination is essential for project success:

Leadership Role: The Team Lead coordinates overall project activities, manages team communication, oversees resource allocation, and tracks progress towards milestones.

Collaboration: Team members collaborate actively on development, testing, documentation, code review, and integration activities.

Communication: The team maintains regular communication about pending tasks, blockers, completed work, and emerging issues.

Decision Making: Project decisions are discussed among team members and recorded appropriately for future reference.

Integration Review: The team conducts integration reviews to verify that individual modules work together correctly before final demonstration.

17. Overall Project Execution Flow

Project Execution Flow

Planning

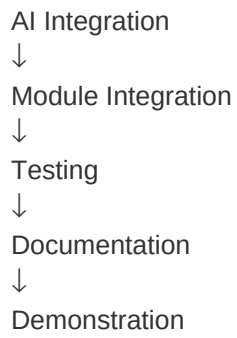


Design



Development





This flow represents the overall progression of the EduGenie project from initial planning through final demonstration, ensuring systematic development and validation of all components.

18. Conclusion

The Project Planning phase establishes a comprehensive, structured approach for developing EduGenie – a Google Gemini-powered learning assistant. This planning document defines the essential framework for project execution, including:

- **Team Organization:** A dedicated team of five members with clear leadership and collaborative roles.
- **Work Breakdown Structure:** Systematic division of work into manageable components across seven major stages.
- **Project Activities:** Detailed activities for each development stage from initiation through demonstration.
- **Technology Resources:** Specification of software tools, frameworks, and APIs required for implementation.
- **Timeline and Milestones:** Sequential phases with clear milestone achievements to track progress.
- **Risk Management:** Identification of potential risks and documented mitigation strategies.
- **Dependencies and Quality:** Clear identification of component dependencies and quality assurance standards.
- **Deliverables:** Eight phases of deliverables aligned with the SmartBridge methodology.

This comprehensive planning provides a structured foundation for the Development phase and ensures that all team members understand project objectives, their responsibilities, and expected outcomes. The plan serves as a reference guide for tracking progress, managing risks, and coordinating team activities throughout the project lifecycle. With this planning framework in place, the EduGenie project is prepared to proceed into systematic development with clear direction, defined resources, and measurable milestones for success.